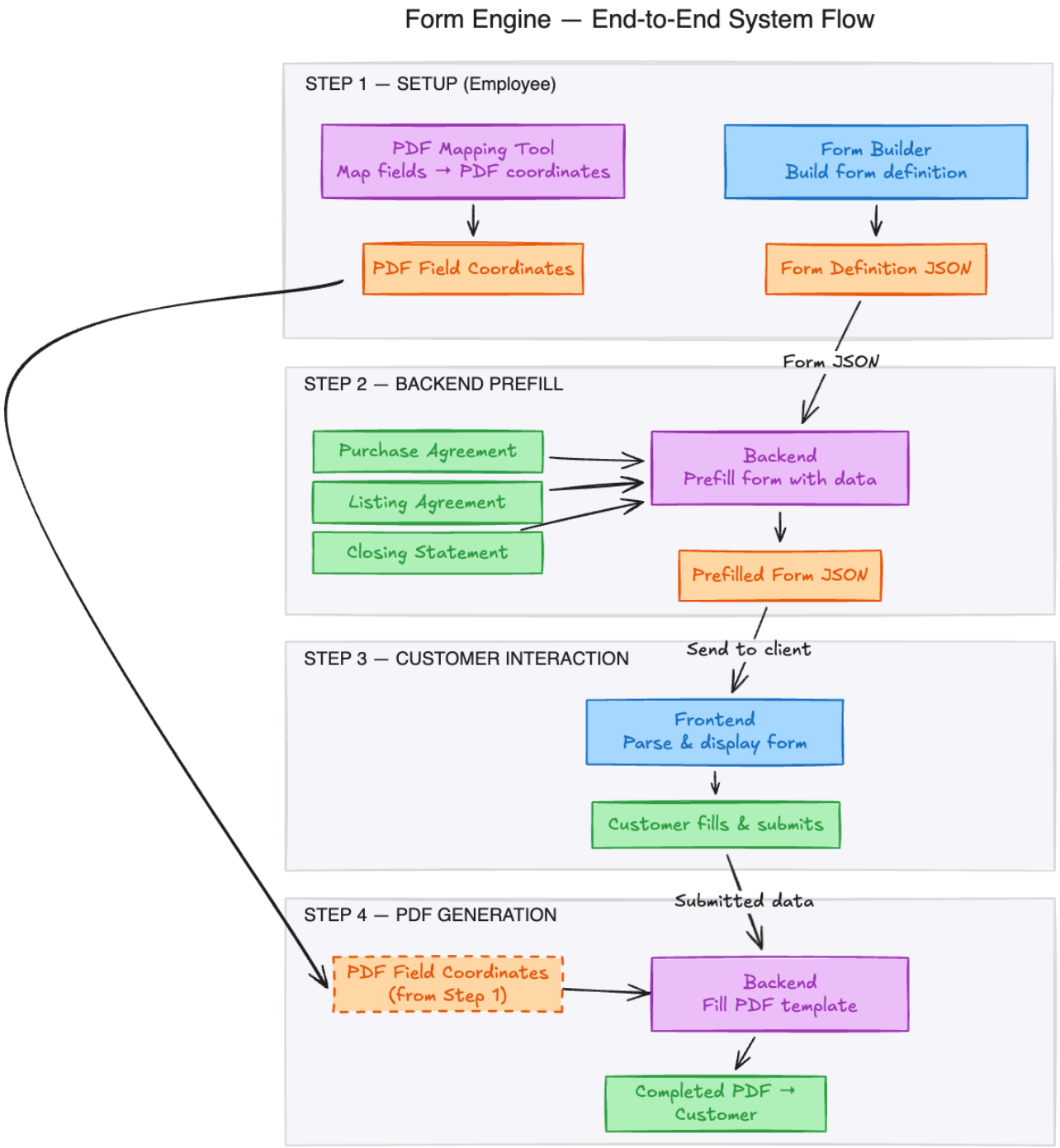
Form Engine — System Design Document

1. System Overview

The Form Engine is a data-driven system for building, prefilling, rendering, and processing dynamic multi-step forms that ultimately produce completed PDF documents. It replaces hand-coded forms with a JSON-driven approach, enabling non-developers to create and modify forms without code changes.

End-to-End Flow



The system involves **four major steps** spanning three actors: the **Employee** (setup), the **Backend** (processing), and the **Customer** (interaction).

Step	Actor	Input	Output
1. PDF Mapping	Employee	Blank PDF template	PDF Field Coordinates
2. Form Builder	Employee	Business requirements	Form Definition JSON
3. Backend Prefill	Backend	Form JSON + data sources	Prefilled Form JSON
4. Customer Interaction	Customer (Frontend)	Prefilled Form	Submitted form data
5. PDF Generation	Backend	Coordinates + submitted data	Completed PDF

---

2. Step 1 — PDF Mapping (Employee)

Purpose

Before any form can generate a PDF, someone must define *where* each piece of data goes on the PDF template. This is the PDF Mapping step.

How It Works

1. The employee opens the **PDF Mapping Tool** — a separate internal application.
2. They upload or select the **blank PDF template** (e.g., a Purchase Agreement PDF, a Listing Agreement PDF).
3. Using the tool's visual interface, they **click on locations** in the PDF where form data should be placed.
4. For each location, they assign a **field ID** that corresponds to a field in the form definition.
5. The tool outputs a **PDF Field Coordinates** mapping — a data structure that links each field ID to its `(x, y)` position, page number, font size, and other rendering properties on the PDF.

Output: PDF Field Coordinates

```
{
  "templateId": "purchase-agreement-v3",
  "fields": [
    {
      "fieldId": "buyerFirstName",
      "page": 1,
      "x": 150,
      "y": 320,
      "fontSize": 12,
      "maxWidth": 200
    },
    {
      "fieldId": "vesselPrice",
      "page": 2,
      "x": 400,
      "y": 185,
      "fontSize": 12,
      "format": "currency"
    }
  ]
}
```

This mapping is stored in the database and referenced later during PDF generation (Step 5).

---

3. Step 2 — Form Builder (Employee)

Purpose

The employee designs the actual form that the customer will interact with. The Form Builder outputs a **Form Definition JSON** — a complete, serializable description of a multi-step form.

How It Works

1. The employee opens the **Form Builder** application.
2. They create **steps** (pages/sections) for the form — e.g., "Vessel Information", "Buyer Details", "Financial Terms".
3. Within each step, they add **fields** by choosing from available field types and configuring their properties.
4. For each field, they configure:
    - **Label & placeholder** — what the customer sees
    - **Validation rules** — required, min/max length, email format, etc.
    - **Data mapping** — where to read prefill data from, and where to write submitted data to
    - **Layout** — column position and span within a responsive grid
    - **Conditional logic** — show/hide fields based on other field values
5. The builder outputs a **Form Definition JSON** that is stored in the database.

Supported Field Types

Type	Description	Example
`text`	Standard text input	First Name, Address
`maskedText`	Formatted text input	Phone number (+1 (###) ###-####)
`number`	Numeric input	Price, Year
`email`	Email with validation	Contact email
`url`	URL with validation	Website
`selectSimple`	Dropdown (string value)	Year, Country
`selectObject`	Dropdown (object value with id+label)	Engine Type, Hull Material
`yachtMake`	Specialized make selector	Vessel make
`yachtModel`	Specialized model selector (depends on make)	Vessel model
`array`	Repeatable group of fields	Deposits, Co-buyers
`conditional`	Toggle or select that shows/hides branches	"Has trade-in?" → show trade-in fields

Output: Form Definition JSON (simplified)

```
{
  "id": "form_purchase_agreement",
  "version": 1,
  "title": "Purchase Agreement",
  "steps": [
    {
```

```
      "id": "step_vessel_info",
      "name": "vessel_info",
      "title": "Vessel Information",
      "order": 0,
      "fields": [
        {
          "type": "text",
          "name": "firstName",
          "label": "First Name",
          "layout": { "column": 1, "colSpan": 6 },
          "dataMapping": {
            "prefillPath": "buyer.firstName",
            "submitPath": "buyer.firstName"
          },
          "validationSchema": [
            ["yup.string"],
            ["yup.required", "First name is required"],
            ["yup.max", 60]
          ]
        }
      ]
    }
  ]
}
```

Key Design Decisions

- **Validation is defined as yup-ast** — JSON-serializable arrays that represent Yup validation schemas. This avoids the limitation of Zod's `refine` / `superRefine` not being serializable.
- **Conditional fields use flat naming** — child fields are siblings, not nested under the conditional's name, avoiding conflicts between the conditional's value (a string/boolean) and its children.
- **Selects support both static options and dynamic API endpoints** with optional infinite scroll for large datasets.

---

4. Step 3 — Backend Prefill

Purpose

Before sending the form to the customer, the backend **prefills** known field values from existing data sources. This saves the customer from re-entering information that the system already has.

Data Sources

The backend pulls prefill data from multiple sources depending on the form type:

Data Source	Examples of Data
Purchase Agreement	Vessel details, buyer info, broker info, price
Listing Agreement	Listing price, seller info, vessel specs
Closing Statement	Financial terms, deposits, adjustments
User Profile	Name, email, phone, address
Vessel Database	Make, model, year, HIN, specifications

How It Works

1. The backend loads the **Form Definition JSON** from the database.
2. For each field in the form, it reads the `dataMapping.prefillPath` property.
3. It looks up that path in the combined data from the relevant sources.
4. If a value is found, it sets that as the field's `defaultValue`.
5. The resulting **Prefilled Form JSON** is sent to the frontend.

Example: Prefill in Action

Given a field definition:

```
{
  "type": "email",
  "name": "buyerEmail",
  "label": "Buyer Email",
  "dataMapping": {
    "prefillPath": "buyer.contactInfo.email",
    "submitPath": "buyerInfo.email"
  }
}
```

And a data source containing:

```
{
  "buyer": {
    "contactInfo": {
      "email": "john.doe@example.com"
    }
  }
}
```

The backend sets `defaultValue: "john.doe@example.com"` on that field before sending the form to the frontend. The customer sees the email pre-filled and can modify it if needed.

---

5. Step 4 — Customer Interaction (Frontend)

Purpose

The frontend application receives the prefilled Form Definition JSON, **parses** it at runtime, and **renders** a fully interactive multi-step form for the customer to complete.

How It Works

1. **Parse the JSON** — The frontend reads the form definition and constructs the UI dynamically.
2. **Build validation schemas** — For each step, the runtime:
   - Reads each field's `validationSchema` (yup-ast arrays)
   - Converts them to real Yup schemas using `@demvsystems/yup-ast`
   - For conditional fields, wraps branch children in `yup.when()` so validation is only active for the visible branch
   - Assembles into a step-level `yup.object().shape({...})`
   - Passes to `yupResolver` for `react-hook-form`

3. **Render fields** — Each field type maps to a specific React component (e.g., `text` → `FormTextField`, `selectObject` → `FormSelect`).
4. **Prefilled values** — Fields with `defaultValue` (set during prefill) appear pre-populated. The customer can review and modify them.
5. **Multi-step navigation** — The customer moves through steps, with per-step validation ensuring data integrity before advancing.
6. **Submit** — On final submission, the frontend collects all field values and maps them using `dataMapping.submitPath` to construct the outgoing payload. This is sent to the backend.

Validation Example

A field defined as:

```
"validationSchema": [
  ["yup.string"],
  ["yup.required", "Email is required"],
  ["yup.email", "Must be a valid email"]
]
```

Gets converted at runtime to:

```
yup.string().required("Email is required").email("Must be a valid email")
```

Conditional Validation

For a conditional field like "Has trade-in?":

```
tradeInValue: yup.when("hasTradeIn", {
  is: true,
  then: yup.number().required("Trade-in value is required").min(1),
  otherwise: yup.mixed().nullable().optional()
})
```

Only when the customer toggles "Yes" does the trade-in value become required.

---

6. Step 5 — PDF Generation (Backend)

Purpose

After the customer submits the form, the backend combines the **submitted data** with the **PDF Field Coordinates** (from Step 1) to produce a **completed PDF**.

How It Works

1. The backend receives the **submitted form data** from the frontend.
2. It retrieves the **PDF Field Coordinates** mapping for the relevant PDF template (created in Step 1).
3. It loads the **blank PDF template** file.
4. For each coordinate entry, it:
   - Looks up the corresponding field value from the submitted data
   - Formats the value if needed (e.g., currency formatting, date formatting)
   - Writes the value onto the PDF at the specified `(x, y)` position, page, and font size
5. The completed PDF is saved and delivered to the **customer** (via download, email, or in-app).

Example

Given submitted data:

```
{
  "buyer": {
    "firstName": "John"
  }
}
```

And PDF coordinates:

```
{
  "fieldId": "buyerFirstName",
  "page": 1,
  "x": 150,
  "y": 320,
  "fontSize": 12
}
```

The system writes "John" at position (150, 320) on page 1 of the PDF template.

---

7. Technical Details

JSON Schema Structure

```
FormSchema
├── id, version, title
└── steps[]
    ├── id, name, title, order
    └── fields[]
        ├── id, type, name, label
        ├── layout: { column, colSpan, order }
        ├── dataMapping: { prefillPath, submitPath }
        ├── validationSchema: YupAst[]
        └── [type-specific properties]
```

Validation: yup-ast

The system uses **yup-ast** (`@demvsystems/yup-ast`) to represent validation rules as JSON arrays. This is critical because:

- **Zod's `refine` / `superRefine`** cannot be serialized to JSON (they require JavaScript functions)
- **Yup's `.when()`** handles conditional validation natively and can be constructed from data
- **yup-ast** provides `transformAll()` to convert JSON arrays → real Yup schema objects

Each yup-ast node is a tuple: `[methodName, ...args]`

Scenario	yup-ast
Required string, 2-60 chars	`[["yup.string"], ["yup.required", "..."], ["yup.min", 2], ["yup.max", 60]]`
Required number	`[["yup.number"], ["yup.required", "..."]]`

Required email	`[["yup.string"], ["yup.required", "..."], ["yup.email", "..."]]`
Optional URL	`[["yup.string"], ["yup.url", "..."], ["yup.nullable"]]`
Object (select)	`[["yup.object"], ["yup.shape", {"id": [...], "label": [...]}], ["yup.required"]]`

Data Mapping

Every field has a `dataMapping` object with two paths:

- **`prefillPath`** — Dot-notation path into the source data for pre-filling (e.g., `"buyer.contactInfo.email"`)
- **`submitPath`** — Dot-notation path for constructing the outgoing payload (e.g., `"buyerInfo.email"`)

This decouples the form's internal field names from both the source data structure and the submission API contract.

Dependencies

Package	Purpose
`react-hook-form`	Form state management
`yup`	Runtime validation
`@demvsystems/yup-ast`	JSON → Yup schema conversion
`@hookform/resolvers`	`yupResolver` bridge